Practical 6 - Understanding T-Test, ANOVA, Chi-Square

Problem - **Q1.** T-Test: Using the teachers' rating data set, does gender affect teaching evaluation rates?

Q2. ANOVA: Using the teachers' rating data set, does beauty score for instructors differ by age?

Q3. Chi-square: Using the teachers' rating data set, is there an association between tenure and gender?

Q4. Correlation: Using the teachers rating dataset, Is teaching evaluation score correlated with beauty score?

Solution - Here is the complete code

Import libraries

```
import pandas as pd
from scipy.stats import ttest_ind
```

Upload the dataset

```
df = pd.read_csv('/content/teachingratings_extracted.csv')
```

```
df.head()
```

	minority	age	gender	credits	beauty	eval	division	native	tenure	students	allstudents	prof	PrimaryLast	vismin	f
0	yes	36	female	more	0.289916	4.3	upper	yes	yes	24	43	1	0	1	
1	yes	36	female	more	0.289916	3.7	upper	yes	yes	86	125	1	0	1	
2	yes	36	female	more	0.289916	3.6	upper	yes	yes	76	125	1	0	1	
3	yes	36	female	more	0.289916	4.4	upper	yes	yes	77	123	1	1	1	
4	no	59	male	more	-0.737732	4.5	upper	yes	yes	17	20	2	0	0	

Next steps: Generate code with df | New interactive sheet

Q1. T-Test: Using the teachers' rating data set, does gender affect teaching evaluation rates?

```
# Separate evaluation scores by gender
male_eval = df[df["gender"] == "male"]["eval"]
female_eval = df[df["gender"] == "female"]["eval"]

# Perform independent two-sample t-test
t_stat, p_value = ttest_ind(
    male_eval,
    female_eval,
    equal_var=False
```

```
)


print("Male mean evaluation =", male_eval.mean())
print("Female mean evaluation =", female_eval.mean())
print("T-statistic =", t_stat)
print("P-value =", p_value)


alpha = 0.05

if p_value < alpha:
    print("Reject H0")
    print("Conclusion: Gender has a significant effect on teaching evaluation rates.")
else:
    print("Fail to reject H0")
    print("Conclusion: There is no significant difference in teaching evaluation rates by gender.")
```

```
Male mean evaluation = 4.069961976642391
Female mean evaluation = 3.901025635156876
T-statistic = 3.2799470144962344
P-value = 0.001123954096040913
Reject H0
Conclusion: Gender has a significant effect on teaching evaluation rates.
```

Interpretations

The average teaching evaluation is higher for male instructors (4.07) than for female instructors (3.90). Since the p-value (0.0011) is less than 0.05, we reject the null hypothesis. Therefore, there is a statistically significant difference between the teaching evaluation ratings of male and female instructors.

Q2. ANOVA: Using the teachers' rating data set, does beauty score for instructors differ by age?

```
from scipy.stats import f_oneway

def age_group(age):
    if age < 40:
        return "Young"
    elif age < 50:
        return "Middle-aged"
    else:
```

```
        return "Older"

df["age_group"] = df["age"].apply(age_group)

young = df[df["age_group"] == "Young"]["beauty"]
middle = df[df["age_group"] == "Middle-aged"]["beauty"]
older = df[df["age_group"] == "Older"]["beauty"]

# Perform one-way ANOVA
F_stat, p_value = f_oneway(young, middle, older)


print("Young mean beauty =", young.mean())
print("Middle-aged mean beauty =", middle.mean())
print("Older mean beauty =", older.mean())
print("F-statistic =", F_stat)
print("P-value =", p_value)


alpha = 0.05

if p_value < alpha:
    print("Reject H0")
    print("Conclusion: Beauty score differs significantly by age group.")
else:
    print("Fail to reject H0")
    print("Conclusion: There is no significant difference in beauty score by age group.")
```

```
Young mean beauty = 0.4134648146373884
Middle-aged mean beauty = -0.03266660264112017
Older mean beauty = -0.18011296099178883
F-statistic = 22.220391462482986
P-value = 6.201291115813017e-10
Reject H0
Conclusion: Beauty score differs significantly by age group.
```

Interpretation

The average beauty score is highest for the Young group (0.413), followed by the Middle-aged group (-0.033) and the Older group (-0.180). The ANOVA test gives an F-statistic of 22.2204 and a p-value of 6.20×10^{-10}. Since the p-value is less than 0.05, we reject the null hypothesis.Therefore, there is a statistically significant difference in beauty scores among the three age groups. The Young group has the highest average beauty score, while the Older group has the lowest.

Q3. Chi-square: Using the teachers' rating data set, is there an association between tenure and gender?

```
from scipy.stats import chi2_contingency

contingency_table = pd.crosstab(
    df["tenure"],
    df["gender"]
)

print("Contingency Table:")
print(contingency_table)

# Perform Chi-square test
chi2_stat, p_value, dof, expected = chi2_contingency(
    contingency_table
)

print("\nChi-square statistic =", chi2_stat)
print("Degrees of freedom =", dof)
print("P-value =", p_value)

alpha = 0.05

if p_value < alpha:
    print("Reject H0")
    print("Conclusion: There is a significant association between tenure and gender.")
else:
    print("Fail to reject H0")
    print("Conclusion: There is no significant association between tenure and gender.")
```

```
Contingency Table:
gender  female  male
tenure
no          50    50
yes        145   213

Chi-square statistic = 2.5083294406381693
Degrees of freedom = 1
P-value = 0.11324592285178972
Fail to reject H0
Conclusion: There is no significant association between tenure and gender.
```

Interpretation

The contingency table shows that both male and female instructors are distributed across the tenure categories.The Chi-square statistic is 2.5083, with 1 degree of freedom. The p-value is 0.1132, which is greater than the significance level of 0.05. Therefore, we fail to reject the null hypothesis. There is no statistically significant association between tenure and gender. In this dataset, tenure status and gender appear to be independent.

Q4. Correlation: Using the teachers rating dataset, Is teaching evaluation score correlated with beauty score?

```
from scipy.stats import pearsonr

# Calculate Pearson correlation
r, p_value = pearsonr(
    df["eval"],
    df["beauty"]
)

print("Pearson correlation coefficient =", r)
print("P-value =", p_value)

alpha = 0.05

if p_value < alpha:
    print("Reject H0")
    print("Conclusion: There is a significant correlation between evaluation score and beauty score.")
else:
    print("Fail to reject H0")
    print("Conclusion: There is no significant correlation between evaluation score and beauty score.")
```

```
Pearson correlation coefficient = 0.18284732895066688
P-value = 8.297585594594811e-05
Reject H0
Conclusion: There is a significant correlation between evaluation score and beauty score.
```

Interpretation

The Pearson correlation coefficient is 0.1828, indicating a weak positive correlation between teaching evaluation score and beauty score. The p-value is 0.000083, which is less than 0.05. Therefore, we reject the null hypothesis. There is a statistically significant but weak

positive correlation between evaluation score and beauty score. This means that instructors with higher beauty scores tend to have slightly higher teaching evaluation scores. However, correlation does not imply causation.

```
Start coding or generate with AI.
```